

CS 458
Spring 2025: Assignment 5

Due in Canvas Assignment **Tuesday, April 29th, 2025** by 11:59 pm

Task1: True/False Questions (3 Ponits)

1. Digital signatures are primarily designed to provide confidentiality of messages. **False**
2. In conventional signatures, there is a one-to-many relationship between a signature and documents, meaning the same person produces identical signatures on different documents. **True**
3. A digital signature requires both the private key and public key of the signer to be created. **False**
4. Message integrity is achieved with digital signatures because any modification to the message after signing will result in a different signature. **True**
5. The RSA digital signature scheme uses the same key generation process as RSA encryption. **True**
6. Security mechanisms are designed to implement security policies while security services are designed to counter security attacks. **True**
7. Non-repudiation means that the sender of a message cannot deny creating the message. **True**
8. Digital signatures are physically included as part of the document, similar to conventional signatures. **False**
9. The verification of a digital signature requires the private key of the signer. **False**
10. In the RSA signature scheme, the signature is generated by "encrypting" the message with the signer's public key. **False**
11. Digital signatures provide message authentication, message integrity, and non-repudiation but not confidentiality. **True**

12. Existential forgery attacks against RSA digital signatures allow attackers to generate a valid signature for any specific message of their choosing. **False**
13. When verifying an RSA digital signature, if the decrypted signature matches the original message, the signature is considered valid. **True**
14. Padding schemes in digital signatures are used to prevent existential forgery attacks. **True**
15. Digital signatures have a one-to-one relationship between the signature and the message, unlike conventional signatures. **True**
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Task2: Multiple Choice Questions (3 Points)

1. Which of the following is NOT one of the core security services mentioned in the lecture?

- A. Confidentiality B. Message integrity C. Non-repudiation **D. Encryption**

2. What does the term "non-repudiation" refer to in the context of digital signatures?

- A. Ensuring message confidentiality
- B. Ensuring the sender cannot deny sending the message**
- C. Preventing message modification
- D. Verifying the recipient's identity

3. In the RSA digital signature verification process, what is used to verify the signature?

- A. The sender's private key B. The receiver's private key
- C. The sender's public key** D. The receiver's public key

4. Which of the following best describes the relationship between a digital signature and the message it signs?

- A. Many-to-one relationship B. One-to-many relationship
- C. One-to-one relationship** D. Many-to-many relationship

5. What is the main limitation of the existential forgery attack against RSA digital signatures?

- A. The attacker needs the private key
- B. The attacker cannot control the content of the message**
- C. The attack only works on short messages
- D. The attack can be easily detected

6. Which of the following is true about digital signatures compared to conventional signatures?

- A. Digital signatures are more easily duplicated
- B. Digital signatures are physically included in the document
- C. Digital signatures are harder to forge**
- D. Digital signatures require physical characteristics to verify

7. What is the recommended minimum bit length for RSA signature schemes according to the lecture?

- A. 256 bits
- B. 1024 bits
- C. 2048 bits**
- D. 4096 bits

8. Which of the following is NOT mentioned as a digital signature scheme in the lecture?

- A. RSA Digital Signature Scheme
- B. AES Digital Signature Scheme**
- C. ElGamal Digital Signature Scheme
- D. Schnorr Digital Signature Scheme

9. What happens when a signed message is modified during transit?

- A. The signature remains valid
- B. The signature verification will fail**
- C. The message will be automatically corrected
- D. The message will be rejected but the signature remains valid

10. Which of the following is NOT an additional security service mentioned in the lecture?

- A. Access control
- B. Auditing
- C. Password management
- D. Physical security** ▼

11. How is confidentiality achieved when using digital signatures?

- A. It is automatically provided by the digital signature
- B. By using an additional layer of encryption**
- C. By using the private key for encrypting the entire message
- D. Confidentiality cannot be achieved with digital signatures

12. In the RSA signature scheme, what mathematical operation is performed to generate a signature?

- A. $s = x^e \bmod n$
- B. $s = x^d \bmod n$**
- C. $s = x + d \bmod n$
- D. $s = x * d \bmod n$

13. What is the purpose of padding in digital signature schemes?

- A. To increase message length
- B. To decrease computation time
- C. To prevent certain attacks including existential forgery**
- D. To ensure message confidentiality

14. Which key verification property makes RSA signature verification typically faster than signing?

- A. The private key can be shorter than the public key
- B. The public key can use a small exponent e**
- C. The modulus can be smaller for verification
- D. Verification uses symmetric cryptography

15. What distinguishes digital signatures from symmetric key authentication?

- A. Digital signatures provide non-repudiation while symmetric key methods do not**
 - B. Digital signatures are faster to compute
 - C. Digital signatures require less computational resources
 - D. Digital signatures provide better confidentiality
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Task3: RSA Signature and Cryptography (4 points)

We have an RSA public-key cryptosystem with Bob's public key: $(N=143, e=7)$ and Alice's public key: $(N=39, e=5)$. Show your work in detail completely:

Bob wants to sign a message $(M = 3)$, then encrypt it and send it to Alice.

Explain the whole process and calculations, and find all the numbers, such as:

Bob's task:

a. Sign a message and calculate the signature value

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Bob's task:

a) Sign a message & calculate the signature value.

Sol:

Given, Bob's public key: $(N=143, e=7)$

Sign the message

$$\text{Factorize } N = 143 \\ = 11 \times 13$$

$$\text{Now, } \phi(N) \\ = \phi(143) = (11-1)(13-1) \\ = 10 \times 12 \\ = 120$$

Now, compute d ;

$$e \times d \equiv 1 \pmod{\phi(N)} \\ 7 \times d \equiv 1 \pmod{120}$$

$$120 = 17 \times 7 + 1 \\ 1 = 120 - 17 \times 7$$

$$\text{So, } d = -17 \pmod{120} \\ = (17 + 120) = 103$$

Hence, $d = 103$

Now, Signature;

$$S = M^d \pmod{N} = 3^{103} \pmod{143}$$

b. Encrypt signature value and send it to Alice

Simplify using modular expansion:

$$3^5 = 243 \equiv 243 - 1 \times 143 = 100 \pmod{143}$$

$$3^{10} \equiv 100^2 = 10000 \pmod{143}$$

$$10000 \div 143 \approx 69.93$$

$$10000 - 69 \times 143 = 10000 - 9867 = 133 \pmod{143}$$

.....

So, on till we get max.

$$3^{103} \pmod{143} = 16$$

$$S = 16$$

b) Encrypt signature value & send it to Alice.

$$C = S^e \pmod{N}$$

where, $S = 16$, $e = 5$, $N = 39$.

$$C = 16^5 \pmod{39}$$

$$16^5 = 16 \times 16 \times 16 \times 16 \times 16$$

$$\text{Now, } 16^2 = 256$$

$$256 \pmod{39} = 256 - 234 = 22$$

$$16^4 = (16^2)^2 \equiv 22^2 = 484$$

$$484 \pmod{39} = 484 - 468 = 16$$

$$\text{Now, } 16^5 \pmod{39}$$

$$16^5 = 16^4 \times 16 \equiv 16 \times 16 = 256$$

$$256 \pmod{39}; 256 \equiv 22 \pmod{39}$$

$$\text{So, } 16^5 \pmod{39} = 22$$

$$C = 22$$

Alice's task:

c. Decrypt the received encrypted signature

d. Verify the signature and signer identity

c) Decrypt the received encrypted signature.

Alice needs her private key $N = 39 = 3 \times 13$
Now, let's find $\phi(N)$

$$\phi(39) = (3-1)(13-1) = (2)(12) = 24$$

Now, $d = 8$, for $e = 5$

$$5(d) \equiv 1 \pmod{24}$$

so, $d = 5$

$$\text{Decrypt ; } S = c^d \pmod{N} = 22^5 \pmod{39}$$

$$22^2 = 484 \pmod{39} = 16$$

$$22^4 = (22^2)^2 \pmod{39} = 256$$

$$22^5 = 22^4 \times 22 = 16$$

$$\boxed{S = 16}$$

d) Verify the signature & signer identity.

$$\text{Bob sends } (M, S) = 3, 16$$

$$\text{Alice verifies ; } M' = S^e \pmod{N} = 16^5 \pmod{143}$$

$$M' = 16^5 \pmod{143} = 3$$

so, $M = M'$ Hence, verified.